Dessertopia – DBMS Week- 2

Identify the Entities :

The main entities in the system are:

1. Customer
2. Admin
3. Category
4. Product
5. Order
6. Order Item
7. Payment
8. Shipment
9. Review

Attributes of Each Entity :

Customer :

- Customer_ID (**Primary Key**)
- Name
- Email
- Phone_Number
- Password
- Address

Admin :

- Admin_ID (**Primary Key**)
- Name
- Email
- Password

Category :

- Category_ID (**Primary Key**)
- Category_Name
- Description

Product :

- Product_ID (**Primary Key**)
- Product_Name
- Price
- Stock
- Description
- Image
- Category_ID (**Foreign Key**)

Order :

- Order_ID (**Primary Key**)
- Order_Date
- Total_Amount
- Order_Status
- Customer_ID (**Foreign Key**)

Order Item :

- OrderItem_ID (**Primary Key**)
- Quantity
- Price
- Order_ID (**Foreign Key**)
- Product_ID (**Foreign Key**)

Payment :

- Payment_ID (**Primary Key**)
- Payment_Method
- Payment_Status
- Payment_Date
- Order_ID (**Foreign Key**)

Shipment :

- Shipment_ID (**Primary Key**)
- Delivery_Address
- Shipment_Status
- Delivery_Date
- Order_ID (**Foreign Key**)

Review :

- Review_ID (**Primary Key**)
- Rating
- Comment
- Review_Date
- Customer_ID (**Foreign Key**)
- Product_ID (**Foreign Key**)

Relationships :

One-to-One (1:1)

Entity 1	Relationship	Entity 2
Order	has	Payment
Order	has	Shipment

One-to-Many (1:M)

Entity 1	Relationship	Entity 2
Customer	places	Orders
Category	contains	Products
Order	contains	Order Items
Product	appears in	Order Items
Customer	writes	Reviews
Product	receives	Reviews
Admin	manages	Products

Primary Keys :

Table	Primary Key
Customer	Customer_ID
Admin	Admin_ID
Category	Category_ID
Product	Product_ID
Order	Order_ID
Order_Item	OrderItem_ID
Payment	Payment_ID
Shipment	Shipment_ID
Review	Review_ID

Create Table (SQL) :

Customer :

```
CREATE TABLE Customer (
  Customer_ID INT PRIMARY KEY,
  Name VARCHAR(100),
  Email VARCHAR(100),
  Phone_Number VARCHAR(15) ,
  Password VARCHAR(50),
  Address VARCHAR(200)
      );
```

Category :

```
CREATE TABLE Category (
  Category_ID INT PRIMARY KEY,
   Category_Name VARCHAR(100),
  Description VARCHAR(200),
              );
```

Product :

```
CREATE TABLE Product (
   Product_ID INT PRIMARY KEY,
   Product_Name VARCHAR(100),
   Price DECIMAL(10,2),
   Stock INT,
   Description VARCHAR(200),
   Image VARCHAR(200),
   Category_ID INT,
   FOREIGN KEY (Category_ID)
   REFERENCES Category(Category_ID)
);
```

Order :

```
CREATE TABLE Orders (
    Order_ID INT PRIMARY KEY,
    Order_Date DATE,
    Total_Amount DECIMAL(10,2),
    Order_Status VARCHAR(50),
    Customer_ID INT,
    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
);
```

Order Item :

```
CREATE TABLE Order_Item (
    OrderItem_ID INT PRIMARY KEY,
    Quantity INT,
    Price DECIMAL(10,2),
    Order_ID INT,
    Product_ID INT,
    FOREIGN KEY (Order_ID)
    REFERENCES Orders(Order_ID),
    FOREIGN KEY (Product_ID)
    REFERENCES Product(Product_ID)
);
```

Payment :

```
CREATE TABLE Payment (
    Payment_ID INT PRIMARY KEY,
    Payment_Method VARCHAR(50),
    Payment_Status VARCHAR(50),
```

```
    Payment_Date DATE,
    Order_ID INT UNIQUE,
    FOREIGN KEY (Order_ID)
    REFERENCES Orders(Order_ID)
);
```

Shipment :

```
CREATE TABLE Shipment (
    Shipment_ID INT PRIMARY KEY,
    Delivery_Address VARCHAR(200),
    Shipment_Status VARCHAR(50),
    Delivery_Date DATE,
    Order_ID INT UNIQUE,
    FOREIGN KEY (Order_ID)
    REFERENCES Orders(Order_ID)
);
```

Review :

```
CREATE TABLE Review (
    Review_ID INT PRIMARY KEY,
    Rating INT,
    Comment VARCHAR(200),
    Review_Date DATE,
    Customer_ID INT,
    Product_ID INT,
    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID),
    FOREIGN KEY (Product_ID)
    REFERENCES Product(Product_ID)
);
```

Admin :

```
CREATE TABLE Admin (
    Admin_ID INT PRIMARY KEY,
    Name VARCHAR(100),
    Email VARCHAR(100),
    Password VARCHAR(50)
);
```

!!! ***A smarter way to discover, order, and enjoy your favourite desserts !!!***